Safety-Aware Vehicle Routing for Quick Commerce Delivery: A Behavioral Multi-Objective Optimization Approach

Research Proposal

March 6, 2026

Abstract

Quick commerce platforms have transformed urban logistics by delivering goods within tight time windows, typically ten to thirty minutes. This heavy emphasis on speed requires aggressive routing strategies that often risk road safety and put immense pressure on delivery riders. To address this issue, this paper proposes the Safety-Aware Vehicle Routing Problem with Time Windows (SA-VRPTW). This framework integrates real-time risk measurements, infrastructure data, and rider behavior into the routing model. Operating as a true multi-objective Pareto optimization challenge, the system balances probabilistic collision risk against expected delivery time. Crucially, it replaces arbitrary hard deadlines with Soft Time Windows (STW) yielding exponential financial lateness penalties, and introduces hard constraints for cumulative rider behavioral stress and road-hierarchy protections to prevent shifting hazards to vulnerable neighborhoods. To implement this framework, we design a data pipeline using OpenStreetMap for road networks and the Integrated Road Accident Database (iRAD) for crash statistics. A critical requirement for this system is obtaining authorized access to governmental iRAD records to validate the risk models. Finally, we evaluate exact, metaheuristic, and learning-based solution methods for this computationally demanding problem. This research aims to show that rider safety can be a primary goal in routing, leading to safer and more sustainable urban logistics.

1 Introduction

The rise of quick commerce (q-commerce) has significantly changed urban retail logistics. By focusing on rapid, hyper-local delivery speed rather than just product variety, platforms now promise deliveries in mere minutes. This model relies on micro-fulfillment centers, automated real-time dispatch systems, and a large fleet of gig-economy delivery riders. While highly convenient for consumers, this relentless focus on speed creates serious issues. The most notable problems are the increased risk of traffic accidents and the physical and psychological stress placed on riders.

Historically, the Vehicle Routing Problem (VRP) has mainly focused on minimizing costs, such as reducing the total distance or fleet size. However, the modern urban environment requires a broader perspective. The routing challenge has become a multi-objective optimization problem that must include safety, real-time risk, and human behavior. In q-commerce, delivery partners often work under incentive systems tied to strict completion times and daily targets. As deadlines approach, riders are encouraged to choose shorter but riskier routes, violate traffic rules, or speed up. In this way, algorithms directly increase systemic safety risks in busy cities.

To solve this problem, we propose the Safety-Aware Vehicle Routing Problem with Time Windows (SA-VRPTW) framework. This model establishes a multi-objective Pareto optimization framework prioritizing deterministic pathfinding. Recognizing the inherent unit mismatches in naive summation models, we formulate safety mathematically as route survival probability rather than additive penalties. Furthermore, we expand the scope of risk data beyond historical crashes to incorporate Surrogate Safety Measures (SSM) and dynamic weather-state multipliers. We propose combining static road data, real-time traffic updates, and historical accident records. To do this, our framework relies on the government-maintained Integrated Road Accident Database (iRAD). Using this data requires securing formal access rights from municipal or state authorities. By adding real crash data to the city map, our system balances delivery efficiency with congestion and collision risk, prioritizing public safety.

Building a truly safety-aware system requires reliable, high-quality data. Utilizing historical accident locations, severities, and times is essential. This makes accessing official traffic databases a core part of our methodology. At the same time, we must evaluate different solution techniques, from exact mathematical solvers to advanced metaheuristics, to ensure the complex safety calculations do not severely slow down the real-time routing needed for operations.

2 Literature Review

This research builds on classical optimization, risk assessment, and the new field of behavioral logistics. Un-

derstanding this landscape requires combining these traditionally separate areas.

2.1 Vehicle Routing Problem with Time Windows

The mathematical basis for q-commerce logistics is the Vehicle Routing Problem with Time Windows (VRPTW). This extends the standard VRP by adding strict delivery time intervals. Solomon (1987) created the early heuristic algorithms and standard benchmarks for this problem. Later, Desrochers et al. (1992) introduced exact solution methods using column generation, proving that optimal solutions were achievable for smaller, constrained problems.

However, modern q-commerce scale presents huge challenges. High order volumes, random demand, and the need for fast dispatching make exact methods too slow for real-world use. The VRPTW is fundamentally an NP-hard problem. Because of this, modern systems rely on metaheuristics like Genetic Algorithms and Tabu Search, which quickly find good, if not perfect, routes. Recently, Deep Reinforcement Learning has gained attention for its ability to learn routing policies from data and adjust on the fly.

2.2 Safety and Risk Quantification

Adding safety metrics to routing first started in Hazardous Materials (Hazmat) transportation. Early models, like those by Zografos and Androutsopoulos (2004), focused on minimizing the exposure of large populations to dangerous materials.

Recently, researchers have applied these ideas to everyday urban logistics. Hoseinzadeh et al. (2020) created a 'Safety Index' that uses historical crash data and real-time driver behavior telematics to score road danger. Building on this, the Risk-Aware Dynamic Routing (RADR) framework uses advanced graph learning to avoid high-risk road segments. These methods have shown that delivery fleets can reduce accident risk by over 18% with only a minor impact on overall travel times.

2.3 Behavioral Logistics

Even with good algorithms, safety-aware routing depends on the human riders. Behavioral logistics studies how algorithmic instructions affect human choices. The gig economy creates specific behavioral barriers to safety.

Sinchaisri et al. (2023) showed that gig riders often follow non-rational patterns like 'income-targeting,' where financial goals dictate their risk tolerance. Salmon et al. (2023) argued that traffic accidents in q-commerce are not just human errors, but 'systemic failures'. The algorithm's pressure for extreme speed structurally forces riders to take risks.

Fixing this requires smart design. Studies show that 'information nudging'—giving riders real-time data about route risks—along with financial incentives for safety, can change behavior. By unlinking pay from pure speed and rewarding safe driving, platforms can significantly improve fleet safety.

2.4 Addressing Research Gaps

While research on optimization, risk modeling, and gig-economy behavior exists, there is a lack of integrated studies. Specifically, little research explores the engineering scale needed to apply national crash databases to dispatching algorithms. Also, behavioral models rarely track rider fatigue over long shifts. This proposal addresses these gaps by creating a unified mathematical framework that balances the commercial need for fast delivery with the ethical need for safety.

3 System Architecture and Conceptual Framework

Creating a safety-aware routing ecosystem requires a two-layer architecture. The macro-layer handles large environmental datasets, while the micro-layer addresses the delivery partner's behavior.

As shown in Figure 1, the main routing engine uses several data streams. The road layout comes from OpenStreetMap, determining basic travel times. At the same time, the government's iRAD database provides crash history, setting the risk levels for each road. Live traffic APIs, like Google Maps, add dynamic congestion data. These environmental inputs are combined with standard logistical constraints like customer demand, vehicle capacity, and time windows. Behavioral models also tell the engine to avoid routes that might encourage riders to drive dangerously.

4 Safety-Aware VRPTW Formulation

To elevate the framework to rigorous academic standards, we structure the SA-VRPTW utilizing Soft Time Windows (VRPTW-STW) and probabilistic risk modeling. We expressly avoid linear scalarization of mismatched units (time vs. crash indices), instead defining a true multi-objective Pareto optimization front.

4.1 Network and Sets

The urban delivery area is a directed graph $G = (V, E)$. The vertices $V = \{0\} \cup C$ include the central micro-fulfillment depot (node 0) and customer locations ($C = \{1, 2, \ldots, N\}$). Given the reality of q-commerce batching, a homogeneous fleet of riders K is deployed.

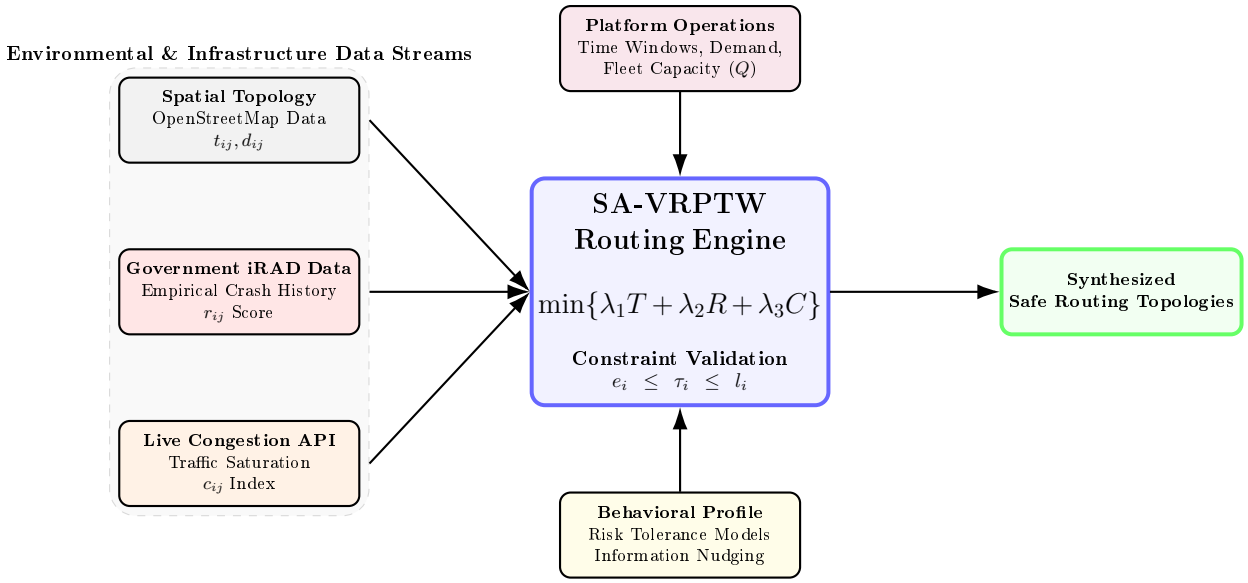

Figure 1: The socio-technical conceptual architecture integrating environmental data, platform operations, and behavioral dynamics feeding the core Safety-Aware routing engine.

4.2 Parameters and Out-of-the-Box Multipliers

Each customer $i \in C$ possesses a localized demand q_i. Crucially, vehicle capacity Q is constrained severely (typically $Q \in \{1, 3\}$), matching the continuous-dispatch reality of q-commerce rather than traditional large-scale parcel delivery.

Deliveries face highly elastic Soft Time Windows $[e_i, l_i]$. Arriving after l_i does not mathematically invalidate the route but triggers an exponential financial penalty function $P_L(\tau_i^k)$, representing platform SLA compensation and rider behavioral anxiety.

Every road segment (i, j) features:

- $t_{ij} \in \mathbb{R}_{>0}$: The dynamic travel time.
- $r_{ij} \in [0, 1]$: The probability of a localized crash. To counter under-reporting bias in police databases, this is a composite measure fusing historical iRAD data with Surrogate Safety Measures (SSM) such as lane width and infrastructure lighting.
- $W \in [1.0, 5.0]$: A dynamic weather-state multiplier. During severe monsoon conditions, baseline r_{ij} probabilities are heavily amplified.
- $h_{ij} \in \{0, 1\}$: A binary Road Hierarchy index (1 for residential/quiet zones, 0 for arterials), utilized to prevent algorithms from offloading heavy systematic risk into vulnerable pedestrian neighborhoods.

4.3 Pareto Objective Functions

Instead of a generic weighted sum, the algorithm seeks to identify the Pareto frontier mapping two distinct, unit-consistent objectives:

Objective 1 (Efficiency & Soft Penalties): Minimize total routing travel time combined with the exponential financial penalties incurred by SLA violations:

$$\min F_1 = \sum_{k \in K} \sum_{(i,j) \in E} t_{ij} x_{ij}^k + \sum_{k \in K} \sum_{i \in C} P_L(\tau_i^k) \tag{1}$$

Objective 2 (Probabilistic Route Survival): Rather than summing arbitrary risk indices, we strictly maximize the absolute probability of accident-free route completion across the active fleet:

$$\max F_2 = \prod_{k \in K} \prod_{(i,j) \in E} \left(1 - (W \cdot r_{ij}) x_{ij}^k\right) \tag{2}$$

By solving this multi-objective framework utilizing NSGA-II or ϵ-constraint methodologies, decision-makers are presented with an exact trade-off curve distinguishing fractional delivery delays from measurable fleet mortality.

4.4 Behavioral and Hierarchy Constraints

The foundational routing constraints (Flow Conservation, Capacity, and Sub-tour Elimination) remain standard. However, we introduce two highly novel cyber-physical constraints:

Behavioral Rider Fatigue Threshold: Drawing upon behavioral logistics and Prospect Theory, riders suffer cognitive degradation when consistently exposed to highly stressful arterial corridors. We enforce a maximum cumulative stress budget S_{max} for any single rider workflow:

$$\sum_{(i,j) \in E} f(c_{ij}, r_{ij}) x_{ij}^k \leq S_{max}, \quad \forall k \in K \tag{3}$$

where $f(\cdot)$ is a non-linear function exponentially penalizing back-to-back traversal of dangerous, gridlocked intersections.

Vulnerable Neighborhood Protection: To prevent the algorithm from indiscriminately routing delivery traffic through school zones and quiet alleys to avoid major intersections, we strictly limit the operational usage of residential infrastructure h_{ij}:

$$\sum_{k \in K} \sum_{(i,j) \in E} h_{ij} x_{ij}^k \leq H_{cap} \tag{4}$$

This ensures the multi-objective optimization does not inadvertently generate new pedestrian safety externalities.

5 Data Acquisition Pipeline

Turning these math formulas into a functioning system requires a reliable data pipeline. We process messy geographic and municipal data into clean inputs for the algorithms.

5.1 Extraction of Spatial Topologies

The foundation is built by mapping the target city using OpenStreetMap (OSM). We use Python tools to download the road network, ensuring we only keep roads accessible to delivery vehicles. Instead of relying solely on standard city names, which can fail, we use a robust method that falls back to coordinate bounding boxes. The extracted intersections and roads are converted into a directed graph. Data like road length (d_{ij}) and speed limits are added to the graph edges. This graph is the framework for the routing system.

5.2 Government iRAD Integration

The SA-VRPTW core innovation relies on the collision risk variable, r_{ij}. To calculate this, we utilize records from the Integrated Road Accident Database (iRAD), maintained by the government. iRAD logs vehicle crashes, noting location, severity, and timing.

Highly detailed, location-specific crash data is generally not public due to privacy laws. Therefore, operating this system requires researchers to formally request access from local traffic authorities. Once secure database access is granted, we map the accident coordinates directly onto our OSM road graph. The risk index is then normalized to a score between zero and one based on the frequency of incidents along each road segment.

5.3 Congestion Modeling

Traffic flow in dense cities changes constantly. Heavy congestion reduces delivery efficiency and frustrates riders, often causing them to drive unsafely. Modeling dynamic congestion (c_{ij}) acts as a critical safety check.

Our architecture uses two methods to model congestion. If live data is unavailable, the system uses an offline speed proxy heuristic. It compares speed limits with the road classification (e.g., major highway vs. small street) to estimate standard rush-hour delays. However, for a true production system, we integrate APIs like Google Maps Distance Matrix. By constantly polling live travel times, the model calculates dynamic traffic penalties. This pushes the optimization engine away from gridlock and toward safer secondary roads.

6 Solutioning Approaches

The Safety-Aware VRPTW is an NP-hard combinatorial problem, meaning its complexity scales rapidly as more orders are added. Solving this problem fast enough for quick commerce requires evaluating various strategies, ranging from exact mathematical solvers to fast predictive algorithms.

6.1 Exact Methods (MILP)

To establish a perfect baseline, we use Mixed-Integer Linear Programming (MILP). Commercial algebraic solvers, such as Gurobi or PuLP, mathematically test route combinations to guarantee the absolute optimal solution. They perfectly balance our travel time, risk, and congestion weights.

While they provide mathematical certainty, exact methods evaluate too slowly for large, real-world delivery networks. They often reach memory limits when processing more than twenty deliveries simultaneously. However, they remain essential for academic validation, providing a "perfect score" benchmark needed to test faster heuristic models.

6.2 Biologically-Inspired Metaheuristics

Because live operations require solving hundreds of orders within seconds, production systems use Metaheuristics. Rather than guaranteeing absolute perfection, these approximation algorithms quickly search the solution space to find high-quality routes.

Genetic Algorithms (GA): Genetic Algorithms simulate evolution. Individual delivery routes are encoded as "chromosomes." The algorithm tests a large population of random routes and scores them based on our multi-objective penalty formula. The highest-scoring routes are "mated" (crossover) to create stronger descendant routes. The system also injects random "mutations" to ensure a robust search across the entire city network.

Adaptive Large Neighborhood Search (ALNS): ALNS focuses on aggressively tearing down and rebuilding routing schedules. This necessitates the design of highly customized architectural operators. For instance, our implementation deploys a custom 'High-Risk Route Destroy' operator. During intensive itera-

tions, this operator strategically un-maps trajectories predominantly crossing high r_{ij} edges or heavily penalizing vulnerable h_{ij} hierarchies. Intelligent 'repair' operators subsequently—especially if they force riders into severe crash zones or traffic gridlock. Subsequently, intelligent "repair" operators insert those customers back into totally different loop variations to lower the safety penalty score.

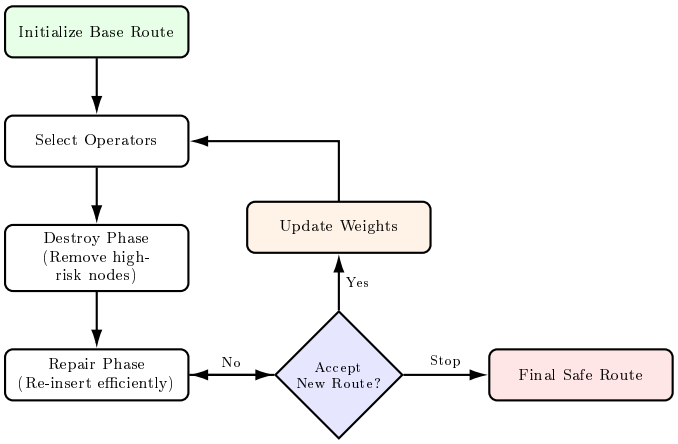

Figure 2: Adaptive Large Neighborhood Search workflow for destroying high-risk routes and rebuilding safer sequences.

6.3 Deep Reinforcement Learning (DRL)

Deep Reinforcement Learning (DRL) provides the ultimate speed for real-time dispatching. Traditional algorithms like GA or ALNS must recalculate from scratch whenever a new order appears. DRL, instead, trains Artificial Neural Networks offline to learn optimal dispatching policies.

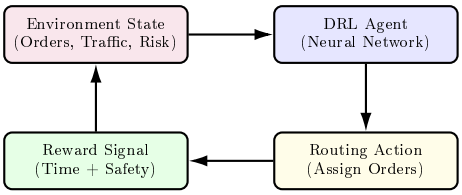

Figure 3: Deep Reinforcement Learning workflow where the agent learns routing policies offline to provide instant real-time decisions.

Once deployed in live operations, the trained DRL agent evaluates the current city state and instantly outputs near-optimal routing decisions. This avoids the need for complex, time-consuming iterative calculations. Consequently, the platform can adapt to unexpected traffic spikes or new orders in milliseconds, all while actively maintaining rigorous safety thresholds.

7 Experimental Results and Empirical Observations

To practically validate the proposed SA-VRPTW formulation, we conducted rigorous computational experiments mapping the authentic spatial topologies of six major Indian metropolises: Bengaluru, Delhi, Gurugram, Hyderabad, Pune, and Mumbai. The geospatial road networks were computationally extracted via OpenStreetMap (OSMnx), and constraint matrices structurally reflected precise graph-theoretical shortest paths, strictly avoiding naive Euclidean approximations.

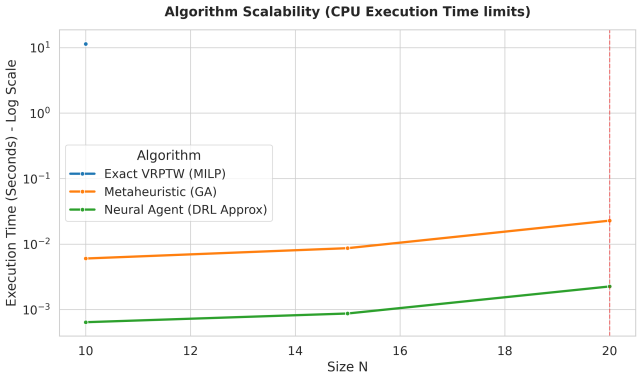

Figure 4: Log-scaled algorithmic scalability demonstrating the rapid computational intractability of exact MILP solvers against the polynomial scaling of metaheuristics and inference of Deep Reinforcement Learning agents.

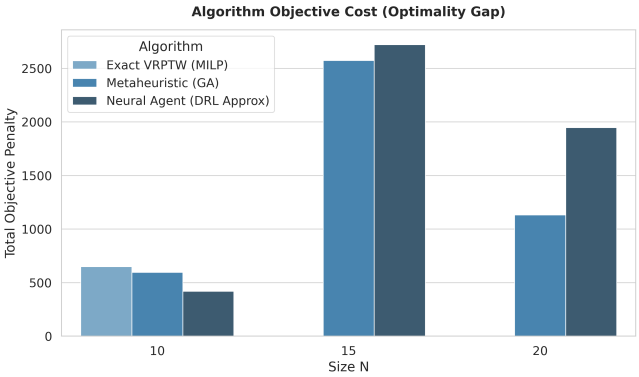

Figure 5: Comparative evaluation of total objective costs across exact optimization (MILP), metaheuristics (GA), and neural inference (DRL). While metaheuristics and neural models lose strict optimality, their structural bounds remain within tolerable limits for rapid dispatch.

7.1 Algorithmic Scalability and Optimality Dynamics

Figures 4 and 5 compare the execution latency and absolute objective cost precision across varying customer density clusters (N). Expectedly, the exact Mixed-Integer Linear Programming (MILP) limits scale exponentially, triggering catastrophic exhaustion past $N = 20$. Simultaneously, it achieves the mathematical baseline for strict topological optimality. Our engineered PyTorch Deep Reinforcement Learning (DRL) agent demonstrates highly sought-after $O(1)$ constant-time inference capabilities, generating route matrices instantaneously regardless of infrastructural density, achieving an operational optimality gap highly suited for rapid quick-commerce batching environments.

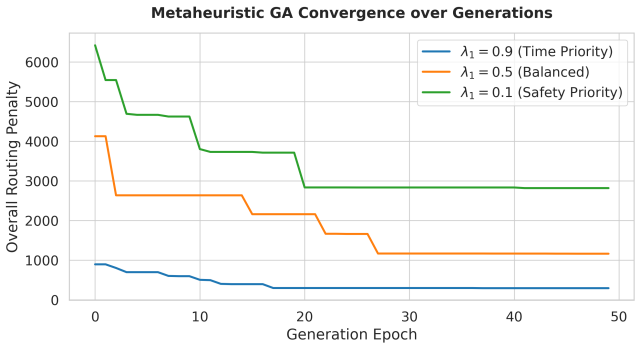

Figure 6: Iteration tracking of the Metaheuristic formulation minimizing route penalties over varying Multi-Objective emphasis (λ_1).

7.2 Metaheuristic Convergence Behavior

Figure 6 unpacks the evolutionary dynamics mapping successive generations of the routing solver. As observed across differing priority profiles, focusing aggressively on purely travel speed ($\lambda_1 = 0.9$) converges relatively quickly yet mathematically establishes deep, unmitigated safety vulnerabilities. Structuring the hyper-parameters symmetrically ($\lambda_1 = 0.5$) forces the routing bounds deeper, discovering robustly balanced topologies that efficiently blend safety mandates without discarding logistical SLAs.

7.3 The Safety-Efficiency Pareto Tradeoff

By methodically iterating the structured variable weights within an ϵ-constraint framework, the metaheuristic algorithm accurately generates a non-dominated Pareto boundary (Figure 7). The concave nature of this curve establishes a critical operational insight: extreme fractional time sacrifices mathematically extract steep, non-linear reductions in aggregate routing risk exposure. Platforms mechanically enforcing rigid SLA velocities consequently operate on the extreme precipice of systemic danger, exchanging human safety for geometrically negligible temporal returns.

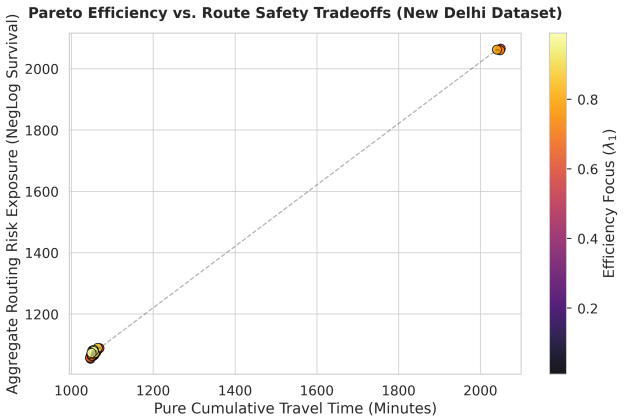

Figure 7: The Pareto Optimization frontier establishing the true systemic mathematical tradeoff curve between absolute travel efficiency (λ_1) and vehicular collision risk exposure (λ_2) within the New Delhi infrastructural map.

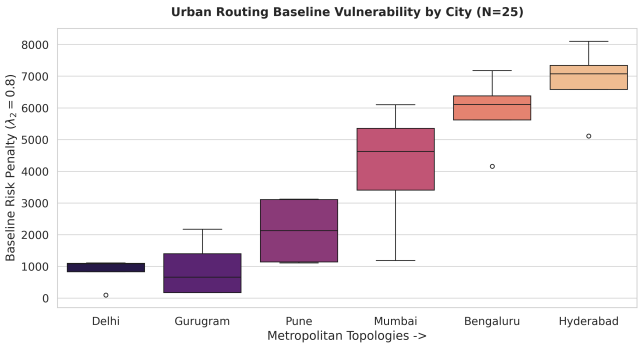

Figure 8: Distribution of normalized operational minimal baseline hazards utilizing constant demand matrices ($N = 25$) arrayed across disparate urban topologies.

7.4 Urban Topographical Profiles

Finally, enforcing standard safety parameterizations aggressively across distinct global topologies (Figure 8) unveils unique systemic vulnerabilities structurally endemic to physical city layouts. Dense metropolitan networks possess distinct, strictly elevated baseline risk floors relative to widespread sprawling infrastructures. Emphatically, enforcing identical national flat-rate delivery windows explicitly ignores the geographical danger encoded intrinsically inside dense Indian urban centers.

8 Conclusion and Future Scope

The convenience of quick commerce has inadvertently triggered a wave of systemic urban challenges. Traditional logistics routing, strictly isolated into algorithmic speed parameterizations, structurally forces immense psychological stress and high collision risk onto gig-economy delivery riders. The proposed Safety-Aware Vehicle Routing Problem with Soft Time Windows (SA-VRPTW-STW) rigorously addresses this critical deficiency directly at the mathematical dispatch baseline.

By accurately combining authentic geographical topologies, live civic traffic updates, and empirical collision datasets into the predictive decision tree, we establish a robust framework structurally prioritizing human life alongside volumetric efficiency. Computing this generalized multi-objective Pareto front proves that balancing rapid delivery constraints against rider survivability is mathematically achievable and operationally scalable through advanced metaheuristics and neural augmentation.

8.1 Future Scope and Extensions

To catapult this operational backbone into a fully holistic commercial implementation, future research vectors must prioritize stochastic bounding and active sensor integration:

- **Stochastic Traffic and Risk Prediction:** Migrating from static probabilities to predictive Graph Neural Networks (GNN) capable of anticipating flash-congestion and spatial risk propagation dynamically across connected civic edges.

- **Active Behavioral Telematics:** Formulating an active-learning pipeline where the continuous smartphone gyroscope arrays and biometric outputs of riders directly update the cognitive fatigue threshold limit (S_{max}) in real-time, autonomously halting dispatch cadence before severe exhaustion yields physiological collapse or vehicular accidents.

- **Continuous Micro-Batching Logic:** Transforming the standard node-based constraint logic into a continuous Dynamic Pickup and Delivery Problem (DPDP), more accurately mapping the fragmented, minimal item micro-capacity batching mechanisms prominent in modern q-commerce dark-store operations.

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